Humberto Bernal, Ph.D.

Economics Researcher · Professor of Applied Microeconomics & Game Theory

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Professional Profile

Applied economist and quantitative researcher with a Ph.D. in Economics from Universidad de los Andes and over 20 years of experience leading empirical research. Proven leadership in directing large-scale data operations, managing multidisciplinary teams, and conducting econometric modeling and policy impact evaluations across academic, governmental, and multilateral institutions, including national ministries and international development banks. Research and teaching expertise bridge Applied Microeconomics, Econometrics, Dynamic Mechanism Design, Game Theory, and Empirical Industrial Organization, leveraging Big Data, Machine Learning, and Agentic AI workflows to address market frictions. Author of research published in leading journals, including Oxford University Press and Taylor & Francis.

Research Pillars & Methodological Toolkit

- **Dynamic Mechanism Design & Game Theory:** Dynamic screening models in unknown environments, asymptotic Bayesian learning, incentive compatibility, contract theory, and adverse selection and moral hazard mitigation.
- **Applied Microeconomics & Industrial Organization:** Causal inference (DiD, Instrumental Variables, RDD), spatial econometrics, firm clustering, external economies of scale, and infrastructure regulation.
- **Big Data, Machine Learning & Computational Stack:** Synthetic cohort pseudo-panels (Deaton framework), statistical matching, micro-imputation, and dynamic simulations. Stack: Python, R, Stata, MATLAB, SQL, Lean 4, Streamlit, and Agentic AI (Antigravity, Cursor).

Featured Job Market Paper

Identifying Rational Types in Unknown Environments under an Indirect Dynamic Mechanism (2026)
Status: Under review. **Interactive Streamlit Simulation** ↗

- Develops an indirect dynamic mechanism proving that a principal's Bayesian beliefs converge almost surely to agents' latent types in unknown environments without direct revelation reports, preventing adverse selection failure.
- Calibrated with microeconometric hazard models and deep learning simulations, supported by an interactive open-source app.

Academic & Professional Experience

Universidad Colegio Mayor de Cundinamarca — **2019 – Present**
Economics Faculty Coordinator & University Professor — *Bogotá, Colombia*

- Lead academic governance and faculty coordination for the Department of Economics, managing over 30 professors.
- Direct curriculum modernization and lead the institutional committee for the renewal of the national certified academic registry (*Registro Calificado*).
- Teach upper-level undergraduate courses in Game Theory and Economic Growth & Development.

G&D Gerencia y Dirección de Proyectos S.A.S. — **June 2022 – Dec 2022**
Project Manager — Ministry of National Education Project — *Bogotá, Colombia*

- Directed the national implementation of the Socio-Occupational Orientation Strategy for Colombia's Ministry of National Education, overseeing project milestones, technical deliverables, and budget execution.
- Supervised multidisciplinary teams of senior analysts and regional coordinators to harmonize secondary and tertiary education pathways with regional labor market demands.
- Structured quantitative monitoring indicators and authored comprehensive strategic policy evaluation reports for government executives.

National Institute for the Deaf (INSOR) — **2019 – 2020**
Senior Economist & Data Desk Coordinator — *Bogotá, Colombia*

- Directed the INSOR Data Desk, supervising econometric modeling and large-scale statistical operations from national household microdata with over 200,000 observations.
- Designed and executed quantitative wage-gap decomposition and labor-access models, identifying structural labor market barriers faced by populations with disabilities.
- Translated empirical findings into national affirmative policy guidelines, culminating in a lead publication in *The Journal of Deaf Studies and Deaf Education* (Oxford University Press, 2025).

Centro Nacional de Consultoría / Contractor to the IDB **2014 – 2018**
Senior External Researcher & Policy Consultant *Bogotá, Colombia*

- Conducted socio-economic, fiscal, and cost-benefit impact evaluations of fourth-generation (4G) highway concessions in Colombia and Nicaragua for the Inter-American Development Bank (IDB) and Centro Nacional de Consultoría.
- Developed econometric frameworks to evaluate toll pricing elasticity, regional trade facilitation, and hold-up risks arising from post-award contract renegotiations in public-private partnerships.

Universidad Externado de Colombia **2008 – 2018**
University Professor of Economics *Bogotá, Colombia*

- Delivered undergraduate and graduate courses in Advanced Microeconomics and Game Theory for cohorts of over 40 students per semester.
- Supervised master's and undergraduate degree theses in applied industrial organization, antitrust economics, and market structure.

Universidad de los Andes & Research Consultancies **2005 – 2015**
Doctoral Researcher & Economic Consultant *Bogotá, Colombia*

- Modeled the regional macroeconomic and fiscal transmission channels of crude oil and natural gas shocks for Universidad de los Andes, estimating price elasticities and local unemployment effects across municipalities.
- Analyzed international commercial competitiveness and tariff advantages across 20 bilateral export commodities for Euromonitor International.
- Prepared technical policy reports and economic evaluations for Fedesarrollo, PROCOLOMBIA, and Fundación FANA.

Education

Ph.D. in Economics, Universidad de los Andes, Bogotá, Colombia **2015**
Specialization in Microeconomic Theory, Industrial Organization, and Econometrics.
Dissertation: Spatial concentration, intra-industry agglomeration, and external economies of scale.

M.Sc. in Economics (Magíster en Economía), Universidad de los Andes, Bogotá, Colombia **2005**

B.S. in Economics (Economista), Universidad Externado de Colombia, Bogotá, Colombia **2003**
Academic Honors: Best Student Scholarship recipient (2000, 2003).

AI Certificates

Certificate of Recognition: Introduction to Agents **November 2026**
Mercor.io Corporation · Credential ID: MRC-2026-CAX2SXAZ **Verify Credential ↗**

- **Course Objective:** Master the architecture, trajectory diagnosis, and evaluation of autonomous AI agents. Focuses on setting up workspace environments, isolating execution failure modes, and iteratively optimizing agent performance through structured feedback loops across context, workspace, and instructions.
- **Professional Application in Economics:** Automates multi-step quantitative workflows, large-scale microdata pipelines, and econometric simulations. Trajectory auditing provides empirical rigor to evaluate automated reasoning, eliminate hallucinations, and enforce strict identification standards across economic modeling.

Selected Peer-Reviewed Publications

- **Bernal, H. (2025).** Empowering deaf rights: Enhancing access to Colombia's labor market. *The Journal of Deaf Studies and Deaf Education* (Oxford Academic), *30*(4), 428–445. doi.org/10.1093/jdsade/enaf034
- **Bernal, H. (2022).** Intra-industry agglomeration and the external economies of scale model: Empirical evidence from Colombia. *Spatial Economic Analysis* (Taylor & Francis), *17*(3), 321–338. doi.org/10.1080/17421772.2021.2009905
- **Bernal, H. (2020).** Demand and Supply in the Cocaine Market: An Empirical Study. *Journal of Globalization and Development*, *11*(1). doi.org/10.1515/jgd-2018-0047
- **Bernal, H. (2019).** Concesiones viales de cuarta generación (4G): Metodología para cuantificar su impacto económico y social. *Revista de Economía & Región*, *13*(2). doi.org/10.32397/er.vol13.n2.1

Languages & Technical Stack

- **Languages:** Spanish (Native); English (Upper-Intermediate / Professional Academic Proficiency, IELTS).
- **Computational Stack:** Python (Pandas, NumPy, Scikit-learn), R, Stata, MATLAB, SQL, Lean 4, Streamlit, LaTeX, Power BI.